Neural Network Verification with Marabou:

Local Robustness of an MNIST Classifier

Formal Verification · SMT-based Analysis · L∞ Robustness

Abstract

We apply the Marabou SMT-based verifier to certify local L∞ robustness of a fully-connected MNIST classifier across 100 test samples and a full ε sweep from 0.01 to 0.20. A two-phase experimental design first identifies SAT samples at ε = 0.05 (3 of 100), then compares robustness trajectories across a robust baseline and two vulnerable samples. Results expose sample-heterogeneous robustness, the SAT–UNSAT runtime asymmetry intrinsic to complete verification, and the critical correctness role of physical pixel-bound clamping.

1. MODEL, DATASET, AND VERIFICATION QUERY

A fully-connected network (784→64→32→10, ReLU activations, ~55,000 parameters) was trained on MNIST for 5 epochs using Adam and cross-entropy loss, achieving ~97.5% test accuracy. The model was exported to ONNX (opset 11) for Marabou's Python API. Architecture size was deliberately constrained: Marabou's complete SMT-based verifier scales exponentially with ReLU neuron count, so a 96-neuron hidden stack (64+32) is a practical upper bound for sub-minute queries at moderate ε.

The verification query is local robustness. Given a correctly-classified test image x with true label d, prove that no perturbation within an L∞ ball of radius ε can alter the predicted class:

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Formally:  ∀x′ s.t. ‖x′ − x‖∞ ≤ ε   ⟹   argmax f(x′) = d
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Marabou encodes the negation as a SAT query — searching for x′ inside the ε-ball where some class j ≠ d outscores d — and reports UNSAT (proof of robustness) or SAT (concrete adversarial example). Input bounds were clamped to the physically valid normalized pixel range [−0.4242, 2.8215]; omitting this clamp causes False SAT at large ε.

Experiments ran in two phases. Phase 1 scanned 100 test images at ε = 0.05 to identify SAT samples. Phase 2 swept ε ∈ [0.01, 0.20] on three representative samples: the all-UNSAT baseline (idx=0, label 7) and two SAT samples (idx=8, label 5; idx=33, label 4) discovered in Phase 1.

2. RESULTS

2.1 Phase 1 — Sample Scan at ε = 0.05

Phase 1 found 3 SAT samples out of 100 at ε = 0.05 (indices 8, 33, 92), confirming that robustness is sample-dependent rather than a universal network property. Three representative samples were selected for Phase 2:

Sample Index	True Label	ε = 0.05 Verdict	Role in Phase 2
idx = 0	7	UNSAT	All-UNSAT baseline
idx = 8	5	SAT → class 6	SAT sample (5→6 confusion)
idx = 33	4	SAT → class 0	SAT sample (4→0 confusion)

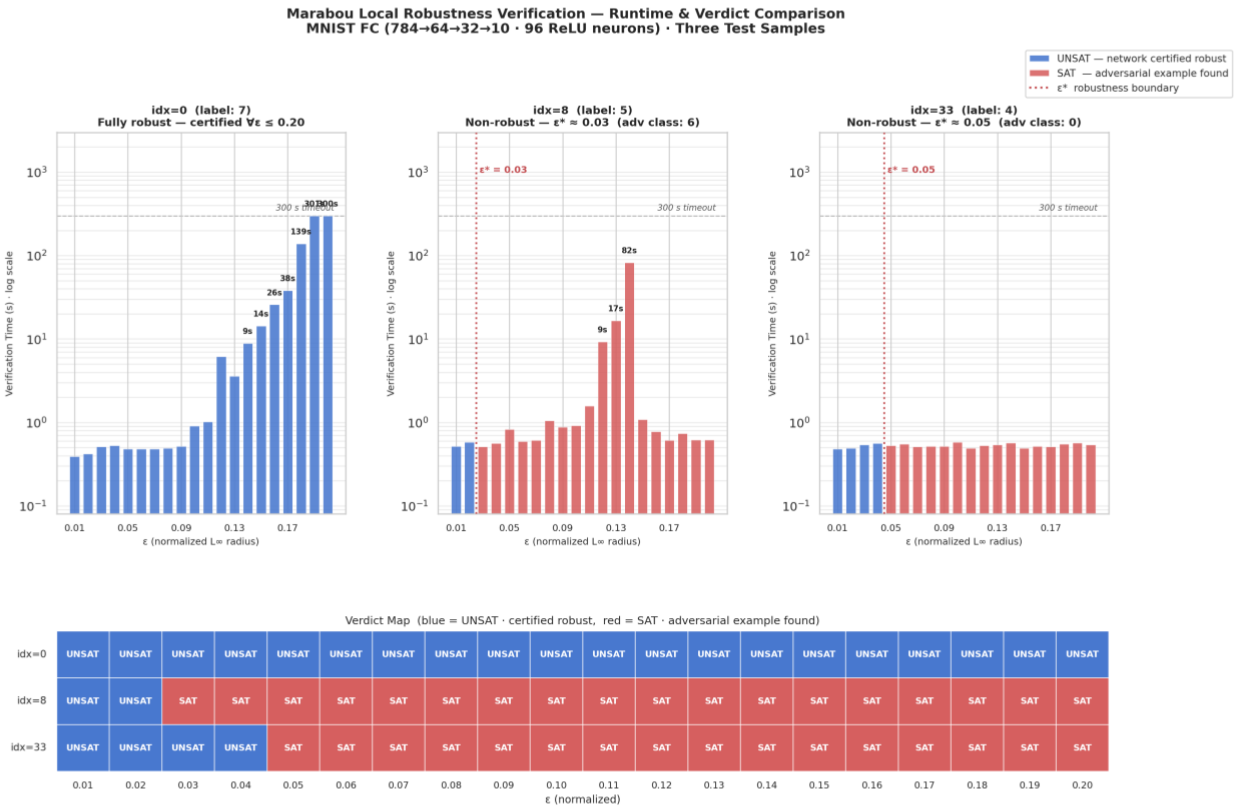
Table 1. Phase 1 scan results at ε = 0.05 and their roles in Phase 2. Adversarial targets: idx=8 → class 6 (5→6 confusion); idx=33 → class 0 (4→0 confusion).

2.2 Phase 2 — Full ε Sweep on Three Representative Samples

Phase 2 produced the following three-way comparison across ε ∈ [0.01, 0.20]. The symbol ✱ marks ε* — the exact robustness boundary where the verdict first transitions to SAT.

ε (norm.)	ε (raw /255)	idx=0 (label 7)	idx=8 (label 5)	idx=33 (label 4)
0.01	~0.8	UNSAT (0.4s)	UNSAT (0.5s)	UNSAT (0.5s)
0.02	~1.6	UNSAT (0.4s)	UNSAT (0.6s)	UNSAT (0.5s)
0.03	~2.4	UNSAT (0.5s)	SAT ✱ (0.5s)	UNSAT (0.5s)
0.05	~4.0	UNSAT (0.5s)	SAT (0.8s)	SAT ✱ (0.5s)
0.10	~8.1	UNSAT (0.9s)	SAT (0.9s)	SAT (0.6s)
0.15	~12.1	UNSAT (14.3s)	SAT (1.1s)	SAT (0.5s)
0.20	~16.2	UNSAT (300.5s)	SAT (0.6s)	SAT (0.5s)

Table 2. Phase 2 full ε sweep. Green = UNSAT (provably robust); red = SAT (adversarial example found). ✱ marks the first SAT transition (ε*).



2.3 Key Findings

Three findings stand out.

First, robustness is sample-heterogeneous. Sample 0 (digit 7) is provably robust across the full sweep — no adversarial input exists even at ε = 0.20 (~16 raw pixel units/255). Samples 8 and 33 have sharp robustness boundaries at ε* ≈ 0.03 and ε* ≈ 0.05 respectively, implying their decision boundaries lie unusually close to the nominal input.

Second, the SAT–UNSAT runtime asymmetry directly reflects the branch-and-bound search structure. UNSAT requires the solver to exhaust the entire 2^{96} case-split tree, producing exponential runtime growth (sub-second for ε ≤ 0.11, then 6s → 14s → 26s → 139s → 300s). SAT terminates on first witness discovery — under 1 second even at ε

= 0.20 — because finding one counterexample is sufficient. This asymmetry is not a Marabou artifact; it is intrinsic to complete verification of an NP-hard problem.

Third, removing physical bounds clamping caused a spurious SAT at $\varepsilon = 0.15$ for sample 0 and inflated $\varepsilon = 0.12$ runtime from 6s to 248s (40× slowdown). The solver explored inputs outside [−0.4242, 2.8215] and found adversarial examples that are physically impossible. Enforcing the valid pixel range eliminates this failure mode entirely.

> ***Remark: Physical Bounds Clamping Is a Correctness Requirement***
>
> Formal verification guarantees soundness with respect to the problem as stated. Without domain knowledge (valid pixel ranges), the SMT solver faithfully searches a region that is mathematically feasible but physically meaningless — producing False SAT. Clamping input bounds to [−0.4242, 2.8215] is not an optimization; it is part of correctly specifying the verification problem.

3. DISCUSSION: STRENGTHS AND LIMITATIONS

3.1 Strengths

(a) Completeness.

An UNSAT result is an unconditional mathematical proof — not a statistical claim or an empirical failure to find an attack. This is qualitatively different from gradient-based methods (FGSM, PGD), which can miss adversarial examples because the attack landscape was not explored in that direction. For safety-critical systems, this distinction is non-negotiable.

(b) Counterexample interpretability.

When the property is violated (SAT), Marabou returns a concrete input assignment with exact logit values, making the failure directly inspectable. This is far more actionable than an attack-based method that reports only a perturbed image without a proof that no smaller perturbation also works.

3.2 Limitations

(a) Scalability.

Even for this 96-neuron network, $\varepsilon = 0.18$ took 139 seconds. The branching factor means worst-case runtime scales as $O(2^n)$, placing networks beyond a few hundred neurons effectively out of reach for complete verification on commodity hardware. This is not a Marabou-specific shortcoming but an intrinsic hardness of the problem (robustness verification is Σ_2^p-complete).

(b) Search space sensitivity.

The false-SAT episode illustrates that the solver is only as reliable as the constraints it is given. Without domain knowledge (valid pixel ranges), the SMT solver faithfully searches a region that is mathematically feasible but physically meaningless. Formal verification guarantees soundness with respect to the problem as stated — the burden of correctly stating the problem is entirely the user's.

(c) Activation function support.

Complete verification requires piecewise-linear activations. Networks using Sigmoid, GELU, or layer normalization require abstraction-based relaxations that trade completeness for tractability, limiting Marabou's applicability to modern transformer-based architectures.